FinTech 545 — Midterm Exam

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Conventions. Arithmetic returns. VaR and ES are positive numbers for a loss. Sample variances use $n-1$. Simulated VaR is the empirical α quantile of P&L (class floor/ceil rule), and ES is minus the mean of the outcomes at or below it. Code: `midterm.py`, labelled by problem and part. Full output: `output_run.txt`.

Problem 1 — Pairwise Correlation with Missing Data (25 pts)

> *Using problem1.csv: 50 days of returns for x1–x5, blanks are missing. Calculate the correlations pairwise, and use the full sample of each series to calculate its variance.*

Each series has 36–42 observations, but only 18 rows are complete, so dropping incomplete rows would lose most of the data. Each pair instead uses the rows where both series are present (28 to 36 rows).

1(a) (5) Calculate the pairwise correlation matrix.

For each pair (i, j) I keep the rows where both are observed and compute the sample correlation on those rows.

	x1	x2	x3	x4	x5
x1	1.0000	0.3328	0.4185	0.2553	0.7323
x2	0.3328	1.0000	0.2510	0.2877	0.7875
x3	0.4185	0.2510	1.0000	0.4836	0.6843
x4	0.2553	0.2877	0.4836	1.0000	0.7826
x5	0.7323	0.7875	0.6843	0.7826	1.0000

x5 is highly correlated with all the others (0.68–0.79), while the others are only 0.25–0.48 correlated with each other.

1(b) (3) Calculate the variance of each series, using every observation that series has.

Sample variance ($n-1$) over all non-missing values of each series:

	Observations	Variance	Std. dev.
x1	41	4.0607e-04	0.020151
x2	41	4.8007e-04	0.021910
x3	36	1.5111e-04	0.012293
x4	41	8.4174e-04	0.029013
x5	42	2.7315e-04	0.016527

1(c) (5) Is the correlation matrix positive semi-definite? Show how you know.

A symmetric matrix is PSD if and only if all of its eigenvalues are ≥ 0. The eigenvalues of the pairwise matrix are:

	Eigenvalue
λ_1	3.085416
λ_2	0.823437
λ_3	0.719085
λ_4	0.461340
λ_5	-0.089279

No, it is not PSD. The smallest eigenvalue is -0.0893, far from zero, and a Cholesky factorization fails. For the corresponding eigenvector v, $v^\top Rv < 0$: a portfolio built on those weights would have negative variance, which is impossible.

The reason is that each entry is estimated on a different subset of rows, so the entries need not be consistent with each other. Here x5 is too correlated with x1–x4 given how weakly x1–x4 are correlated among themselves.

1(d) (7) If it is not, use Higham's method to find the nearest positive semi-definite correlation matrix. Report the matrix and its eigenvalues. Which entries changed the most?

Higham's method finds the matrix closest to the pairwise matrix in Frobenius norm, subject to being PSD with a unit diagonal. It alternates two projections: one clips the negative eigenvalues to zero, the other resets the diagonal to 1. A Dykstra correction makes it converge to the *nearest* such matrix.

	x1	x2	x3	x4	x5
x1	1.0000	0.3495	0.4259	0.2708	0.6946
x2	0.3495	1.0000	0.2606	0.3075	0.7393
x3	0.4259	0.2606	1.0000	0.4925	0.6626
x4	0.2708	0.3075	0.4925	1.0000	0.7377
x5	0.6946	0.7393	0.6626	0.7377	1.0000

Eigenvalues:

	Eigenvalue
λ_1	3.039734
λ_2	0.806708
λ_3	0.703476
λ_4	0.450082
λ_5	-5.29e-10 (= 0)

The negative eigenvalue is now zero (the 10^{-10} is rounding). The other eigenvalues barely move, and the Frobenius distance to the original is 0.121.

Changes, largest first:

Pair	Pairwise	Higham	Change
x2–x5	0.7875	0.7393	-0.0482
x4–x5	0.7826	0.7377	-0.0450
x1–x5	0.7323	0.6946	-0.0377
x3–x5	0.6843	0.6626	-0.0217
x2–x4	0.2877	0.3075	+0.0198
x1–x2	0.3328	0.3495	+0.0166
x1–x4	0.2553	0.2708	+0.0155
x2–x3	0.2510	0.2606	+0.0096

Pair	Pairwise	Higham	Change
x3–x4	0.4836	0.4925	+0.0089
x1–x3	0.4185	0.4259	+0.0075

The four largest changes are the four correlations with x5, each lowered by 0.022 to 0.048. All the other correlations rise slightly (0.007 to 0.020). This fits the diagnosis in (c): the x5 row was the inconsistent part.

1(e) (5) Combine your results into a covariance matrix. Report it.

$\Sigma = DRD$, with $D = \text{diag}(\sigma_i)$ from part (b) and R the Higham matrix from (d). I use the Higham matrix because the pairwise one is not PSD, and multiplying by D would not fix that.

	x1	x2	x3	x4	x5
x1	4.0607e-04	1.5429e-04	1.0551e-04	1.5831e-04	2.3132e-04
x2	1.5429e-04	4.8007e-04	7.0189e-05	1.9549e-04	2.6773e-04
x3	1.0551e-04	7.0189e-05	1.5111e-04	1.7565e-04	1.3461e-04
x4	1.5831e-04	1.9549e-04	1.7565e-04	8.4174e-04	3.5372e-04
x5	2.3132e-04	2.6773e-04	1.3461e-04	3.5372e-04	2.7315e-04

Its diagonal equals the part (b) variances, and its eigenvalues are all ≥ 0 (the smallest is ≈ 0), so it is a valid PSD covariance.

Problem 2 — VaR and ES for a Single Stock (20 pts)

> *Using problem2.csv: 501 daily prices for XYZ and SPY. You hold 100 shares of XYZ.* $r_{XYZ} = \alpha + \beta r_{SPY} + \epsilon$. *Assume the arithmetic returns of SPY and* ϵ *are normally distributed.*

Model. From 500 daily returns:

- OLS gives $\hat{\alpha} = -3.04 \times 10^{-4}$ (not significant, $t = -0.47$), $\hat{\beta} = 1.1933$ ($t = 20.0$), $R^2 = 0.446$, and residual s.d. $\hat{\sigma}_\epsilon = 0.014605$.
- SPY: $\hat{\mu}_S = 2.548 \times 10^{-4}$, $\hat{\sigma}_S = 0.010972$.
- Jarque–Bera does not reject normality for SPY ($p = 0.95$) or for the residuals ($p = 0.10$).

Since r_{SPY} and ϵ are independent normals, r_{XYZ} is normal with

$$\mu = \hat{\alpha} + \hat{\beta}\hat{\mu}_S \approx 0, \qquad \sigma = \sqrt{\hat{\beta}^2\hat{\sigma}_S^2 + \hat{\sigma}_\epsilon^2} = 0.019614.$$

The position is worth $100 \times \$74.5183 = \$7{,}451.83$. Because $\mu \approx 0$, absolute and relative VaR are the same.

2(a) (10) What is your VaR at the 95% and 99% levels, in percent? Explain your method.

For a normal return, $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$ with $z_{0.05} = -1.6449$ and $z_{0.01} = -2.3263$.

- **95% VaR = 3.23%** ($1.6449 \times 1.9614\%$), about \$240.42
- **99% VaR = 4.56%** ($2.3263 \times 1.9614\%$), about \$340.03

2(b) (10) What is your ES at the 95% and 99% levels, in percent? Explain your method.

ES is the mean loss beyond VaR. For a normal, $\text{ES}_\alpha = -\mu + \sigma\,\phi(z_\alpha)/\alpha$.

- **95% ES = 4.05%** ($1.9614\% \times 0.10314/0.05$), about \$301.49
- **99% ES = 5.23%** ($1.9614\% \times 0.026652/0.01$), about \$389.56

A 100,000-draw simulation of the same model (r_{SPY} and ϵ drawn separately) gives VaR 3.24% / 4.58% and ES 4.07% / 5.26%, matching the closed form within simulation error.

Problem 3 — Copula VaR and ES for a Two Book Portfolio (75 pts, includes 20 extra credit)

Using problem3.csv and problem3_holdings.csv: 1,001 daily prices for S1–S10, holdings and book (A or B) for each. Value each position at the last price.

1,000 daily returns per stock. At the last price, Book A (S1–S5) is worth \$200,618.44, Book B (S6–S10) \$235,277.92, total **\$435,896.36**. S9 is the largest position (\$98,248.71).

3(a) (25) For each stock, fit a normal, a Student's t, and a normal inverse Gaussian distribution to its returns. Report the fitted parameters. Which distribution fits each stock best? Justify your choice.

All three are fitted by maximum likelihood on the raw (not de-meaned) returns:

- Normal (μ, σ): 2 parameters.
- t (μ, σ, ν): 3 parameters, fat tails, symmetric.
- NIG $(\mu, \alpha, \beta, \delta)$: 4 parameters, fat tails and skew ($\beta < 0$ means a longer left tail).

I choose by **AICc** $= 2k - 2\ln L + \frac{2k(k+1)}{n-k-1}$, which penalizes the extra parameters. BIC gives the same choices.

Normal and t parameters:

Stock	Normal μ	Normal σ	t μ	t σ	t ν
S1	0.0009037	0.01183	0.0009037	0.01182	1.82e+12
S2	0.0008003	0.01535	0.0005814	0.01167	4.858
S3	0.0004236	0.0189	0.002291	0.01366	4.161
S4	0.00078	0.009933	0.00078	0.009928	8.43e+13
S5	0.0007088	0.01895	0.0008333	0.01558	6.289
S6	0.0008047	0.01509	0.002445	0.01065	3.824
S7	0.0006927	0.01367	0.0006927	0.01367	5.35e+12
S8	0.000374	0.01711	0.0006541	0.01142	3.483
S9	0.001746	0.02065	0.003383	0.01531	4.385
S10	0.0001648	0.01306	0.0004047	0.009427	4.121

NIG parameters:

Stock	NIG μ	NIG α	NIG β	NIG δ
S1	0.07656	2086	-588.2	0.2575
S2	-0.0003046	73.21	4.864	0.01659
S3	0.01047	79.84	-36.34	0.01966
S4	0.08955	3368	-985	0.2903
S5	0.001497	73.58	-2.233	0.02596
S6	0.008272	91.68	-41.69	0.01463
S7	0.07748	1636	-443.7	0.2726
S8	0.001183	50.62	-2.887	0.01416
S9	0.01013	63.16	-22.93	0.02151
S10	0.001049	76.69	-5.368	0.0126

Log-likelihood, AICc (lower is better), choice:

Stock	LL normal	LL t	LL NIG	AICc normal	AICc t	AICc NIG	Chosen
S1	3019.00	3019.00	3019.13	-6033.99	-6031.98	-6030.22	Normal
S2	2758.36	2816.69	2815.05	-5512.71	-5627.36	-5622.06	t
S3	2550.26	2622.24	2644.84	-5096.51	-5238.46	-5281.64	NIG
S4	3193.49	3193.49	3193.58	-6382.97	-6380.96	-6379.12	Normal
S5	2547.45	2578.45	2577.12	-5090.89	-5150.87	-5146.21	t
S6	2775.61	2847.38	2872.72	-5547.20	-5688.74	-5737.40	NIG
S7	2873.85	2873.85	2873.96	-5743.69	-5741.68	-5739.89	Normal
S8	2649.56	2749.66	2747.57	-5295.11	-5493.29	-5487.09	t
S9	2461.76	2521.39	2536.77	-4919.51	-5036.76	-5065.51	NIG
S10	2919.67	2990.44	2989.37	-5835.32	-5974.86	-5970.70	t

Justification.

- **S1, S4, S7 → Normal.** Skew and excess kurtosis are about zero. The t fit goes to $\nu \to \infty$ and the NIG to $\alpha \to \infty$, both of which are the normal limit. Their log-likelihood gain (at most 0.13) does not pay for the extra parameters.
- **S2, S5, S8, S10 → t.** These have fat tails (excess kurtosis 2.7–6.5, fitted $\nu \approx 3.5$–6.3) but small skew. The t beats the normal by 60–198 AICc points. The NIG's skew parameter adds nothing (its log-likelihood is even slightly lower), so its AICc is 4–6 worse.
- **S3, S6, S9 → NIG.** These have fat tails *and* strong negative skew (−1.1 to −1.3). The NIG fits a clearly negative β and beats the t by 29–49 AICc points. A symmetric t cannot capture the long left tail, which is the part that matters for VaR/ES.

3(b) (20) Using the distributions you chose in part (a), fit a Gaussian copula and a Student's t copula. Which one fits better? Justify your choice.

Method.

1. Transform each stock's returns to uniforms with its chosen CDF: $u_{t,i} = \hat{F}_i(r_{t,i})$.
2. Estimate R from Kendall's τ, $\rho = \sin(\pi\tau/2)$. This holds for both the Gaussian and the t copula, so both use the same R and the comparison is only about the copula family. R is PD (smallest eigenvalue 0.38).
3. Gaussian copula log-likelihood with $z = \Phi^{-1}(u)$ (no extra parameters). t copula log-likelihood with $y = t_\nu^{-1}(u)$, maximized over $\nu \in (2, 100]$ (one extra parameter).
4. Compare with AICc and BIC ($m = 1{,}000$).

Copula correlation matrix R:

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
S1	1.000	0.543	0.528	0.562	0.542	0.294	0.249	0.232	0.281	0.293
S2	0.543	1.000	0.564	0.530	0.523	0.330	0.233	0.246	0.310	0.276
S3	0.528	0.564	1.000	0.532	0.571	0.316	0.303	0.268	0.340	0.298
S4	0.562	0.530	0.532	1.000	0.568	0.306	0.233	0.269	0.296	0.299
S5	0.542	0.523	0.571	0.568	1.000	0.333	0.270	0.279	0.276	0.296
S6	0.294	0.330	0.316	0.306	0.333	1.000	0.535	0.524	0.539	0.547
S7	0.249	0.233	0.303	0.233	0.270	0.535	1.000	0.526	0.522	0.542
S8	0.232	0.246	0.268	0.269	0.279	0.524	0.526	1.000	0.520	0.524
S9	0.281	0.310	0.340	0.296	0.276	0.539	0.522	0.520	1.000	0.522
S10	0.293	0.276	0.298	0.299	0.296	0.547	0.542	0.524	0.522	1.000

There is a clear block structure: correlations are about 0.52–0.57 within each book but only 0.23–0.34 across the books.

	Gaussian	t
loglik	2,110.02	2,108.16
k (copula params)	0.00	1.00
nu		100.00
AICc	-4,220.04	-4,214.32
BIC	-4,220.04	-4,209.42

The Gaussian copula fits better. The t copula's log-likelihood keeps rising in ν (1594 at $\nu = 4$, 2026 at $\nu = 12$, 2108 at $\nu = 100$) and hits the upper bound. As $\nu \to \infty$ the t copula becomes the Gaussian, so the data show no extra tail dependence. The Gaussian has the higher log-likelihood with one fewer parameter, and it wins on both AICc (-4220.0 vs -4214.3) and BIC. The fat tails and skew are still in the model, through the t and NIG margins.

3(c) (30) Using the copula you chose in part (b), simulate the portfolio with at least 5,000 draws. Calculate the absolute VaR and ES at the 95% level, in dollars, for each stock, each book, and the total portfolio. Present the results in a table. Discuss the results.

Method. Draw 100,000 vectors $Z \sim N(0, R)$ and set $U = \Phi(Z)$. Then map to returns through each stock's chosen quantile function, $r_i = \hat{F}_i^{-1}(U_i)$. The P&L of each position is value × return. Book and total P&L are the sums *within each draw*. Absolute VaR is minus the 5% quantile of P&L (measured from zero), and ES is minus the mean of the worst 5%.

Position	Book	Value ($)	VaR ($)	ES ($)	VaR (% of value)	ES (% of value)
S1	A	48,329.48	899.30	1,136.63	1.86%	2.35%
S2	A	46,697.43	1,081.43	1,579.43	2.32%	3.38%
S3	A	24,267.28	799.91	1,191.25	3.30%	4.91%
S4	A	43,596.81	676.98	860.60	1.55%	1.97%
S5	A	37,727.45	1,098.59	1,536.87	2.91%	4.07%
S6	B	46,326.64	1,216.47	1,855.54	2.63%	4.01%
S7	B	40,144.31	871.21	1,101.62	2.17%	2.74%
S8	B	26,045.88	646.42	1,037.26	2.48%	3.98%
S9	B	98,248.71	3,372.72	5,093.74	3.43%	5.18%
S10	B	24,512.38	477.99	726.90	1.95%	2.97%
Book A	A	200,618.44	3,614.14	4,905.75	1.80%	2.45%
Book B	B	235,277.92	5,405.84	7,850.13	2.30%	3.34%
Total		435,896.36	7,654.43	10,656.56	1.76%	2.44%

Discussion.

- **Stocks.** S9 has by far the most risk (VaR \$3,373, ES \$5,094): it is the largest position, the most volatile stock, and negatively skewed. ES/VaR is about 1.26 for the normal stocks (the normal value), but 1.40–1.60 for the t and NIG stocks, whose losses beyond VaR are much bigger. S3 has about the same value as S8 and S10, but its ES is 15% and 64% higher than theirs because of the skew.
- **Books.** Book B is riskier than Book A (VaR 2.30% vs 1.80% of value). Book B holds S9 and two of the three skewed stocks, while two of Book A's largest positions are normal stocks.
- **Diversification.** Total VaR (\$7,654) is 31.3% below the sum of the stock VaRs (\$11,141.02), and 15.1% below the sum of the two book VaRs. The low cross-book correlation is what makes combining the books worthwhile. Adding up the book numbers would overstate total risk by about 17.8%.
- **Effect of the margins.** Re-running with normal margins for every stock (same copula, same draws) gives a total ES of \$9,546 instead of \$10,657, about 10% lower. For individual stocks the gap ranges from 6% to 32%, largest for S3, S6 and S9. A normal model would understate the tail loss.
- **Accuracy.** Across six seeds, the standard deviation of total VaR and ES is \$41.37 and \$44.45, under 1% of the values. 100,000 draws is enough.